

Fundamentals of Electric Circuits

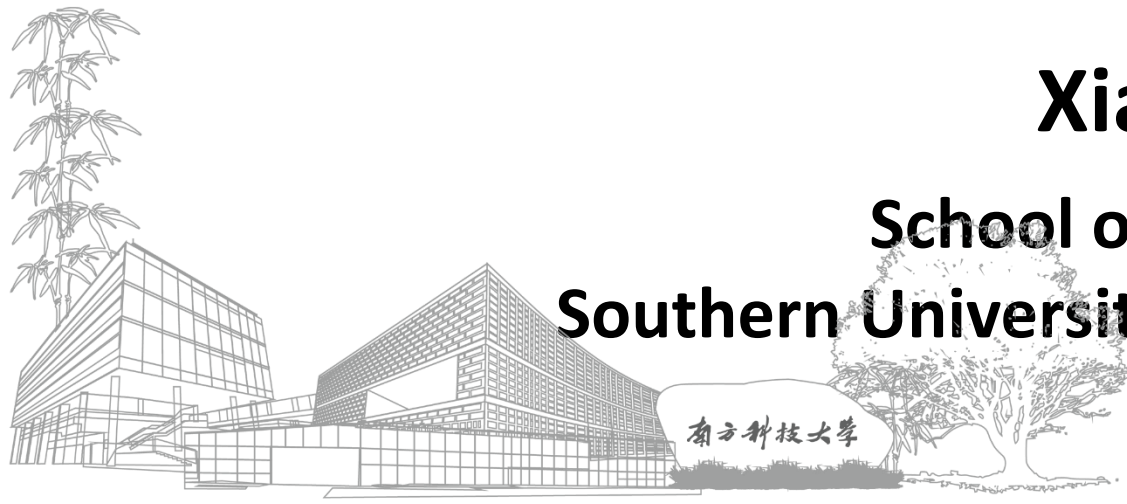
电路基础

Lecture2: Basic Laws

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Outline

- **Ohm's Law**
- **Terminology**
 - **Branches, Nodes, Loops, Meshes**
- **Kirchhoff's Law**
- **Series Resistors and Voltage Division**
- **Parallel Resistors and Current Division**
- **Wye-Delta Transformations**

Resistance

- Different material allow charges to move within them with different levels of ease. This physical property or ability to resist current is known as **resistance** and is represented by the **symbol R** → Resistor (电阻器): A device provides resistance
- The resistance of any material with a uniform **cross-sectional area A** and **length l** is **inversely proportional to A** and **directly proportional to l** . The constant of the proportionality is the **resistivity (电阻率)** of the material, i.e., ρ

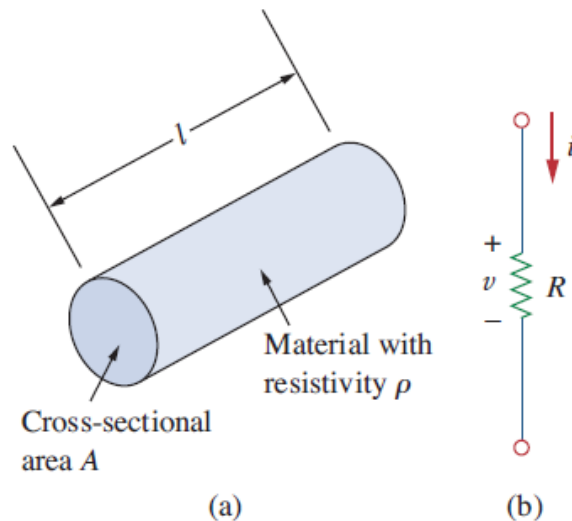


Figure 2.1

(a) Resistor, (b) Circuit symbol for resistance.

$$R = \rho \frac{l}{A}$$

Unit: Ohm (Ω)

TABLE 2.1

Resistivities of common materials.

Material	Resistivity ($\Omega \cdot \text{m}$)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^2	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator

Ohm's law

- **Ohm's law (欧姆定律):** The voltage v across a resistor is directly proportional to the current i flowing through the resistor. The proportionality constant is the resistance of the resistor

$$v(t) = Ri(t)$$

- One can also write

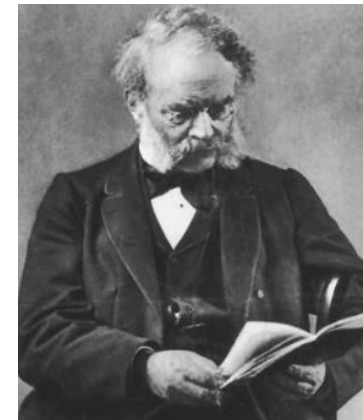
$$i(t) = \frac{1}{R} v(t) \Rightarrow i(t) = Gv(t)$$

- The **conductance G (电导)** is a measure of how well an element can conduct electric current, measured in **Siemens (S)**
- Instantaneous power dissipated in a resistor

$$p(t) = v(t)i(t) = \frac{v^2(t)}{R} = Ri^2(t)$$

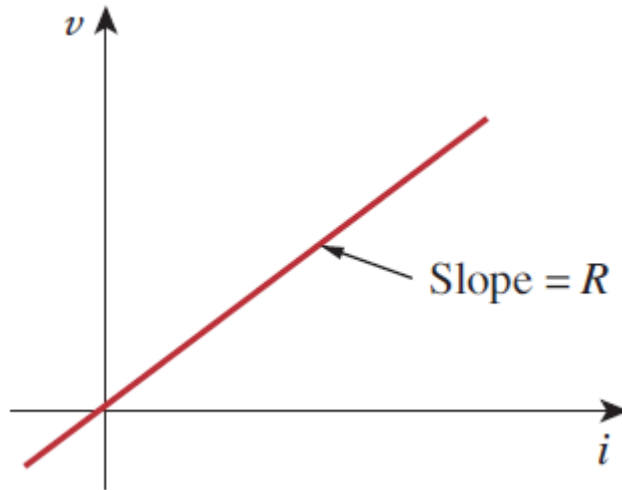


George Simon Ohm
(1787-1854)
German physicist

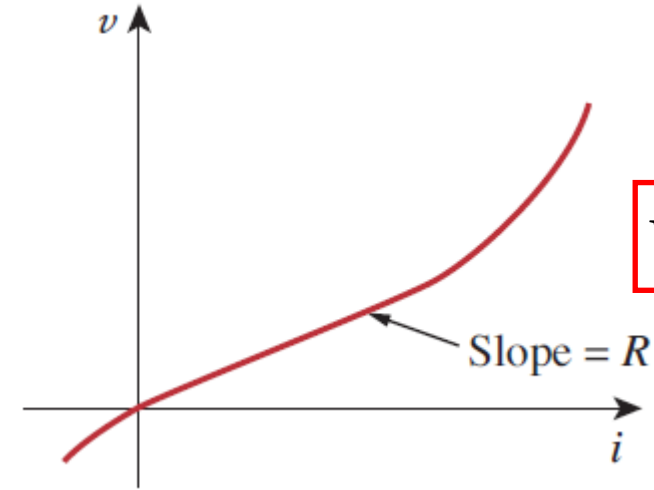


Werner von Siemens
(1816-1892)
German Inventor

Linear and Nonlinear Resistors



Linear resistor



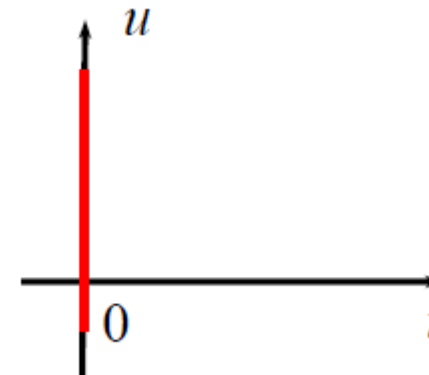
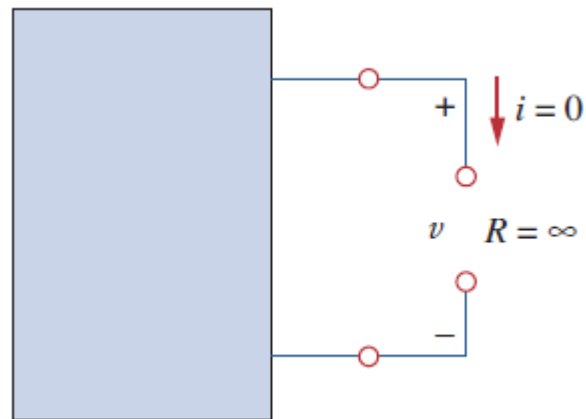
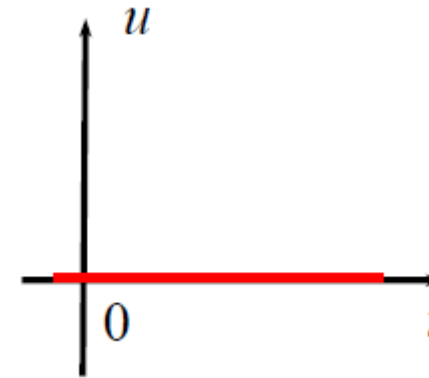
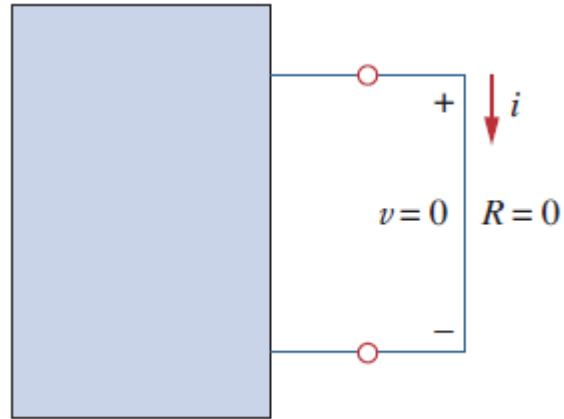
Nonlinear resistor

$$v = R(i) \cdot i$$

- In this course, we assume that all the elements that are designated as resistors are linear (unless mentioned otherwise)
- *Q: Is the resistance of the filament (电阻丝) of a lamp bulb constant? If not, how does it change with temperature?*

Short and Open Circuits

- A device with zero resistance is called **short circuit (短路)** and a device with zero conductance (i.e., infinite resistance) is called **open-circuit (开路)**



Resistors

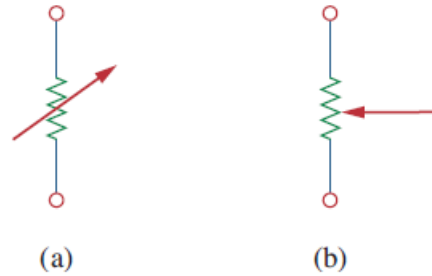


Figure 2.4

Circuit symbol for: (a) a variable resistor in general, (b) a potentiometer.

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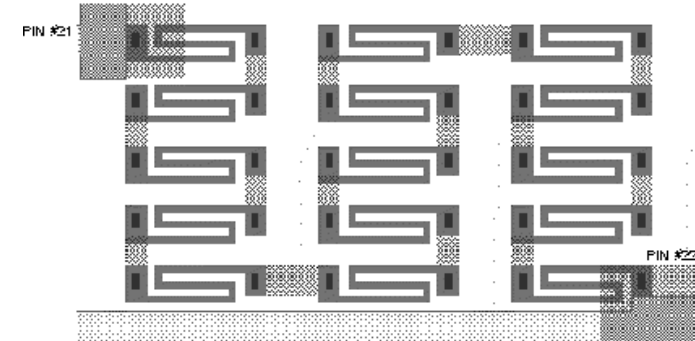
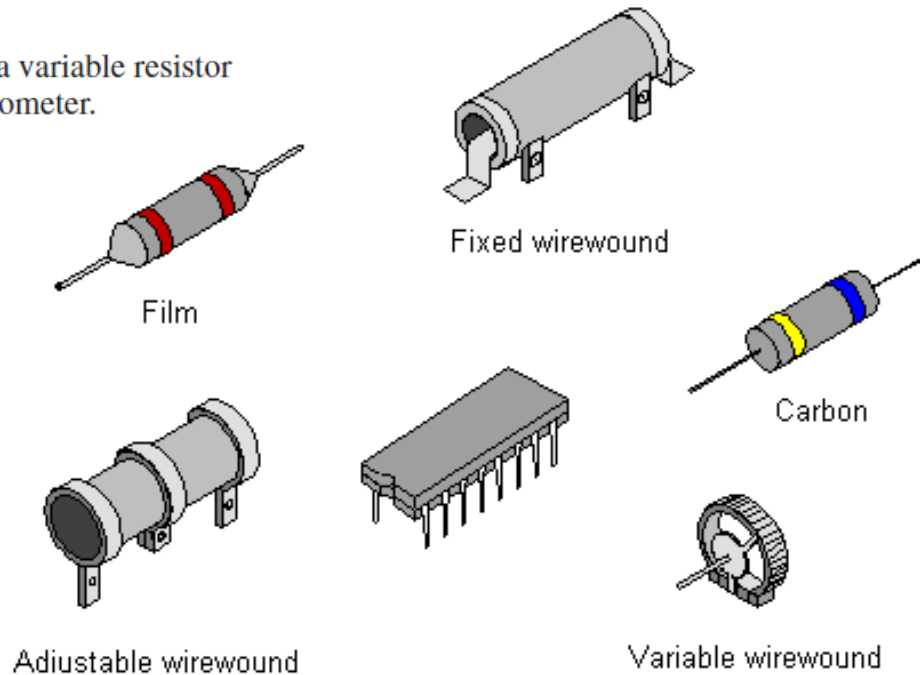


Figure 2.6

Resistors in an integrated circuit board.

Example: Ohm's Law

Example 2.2

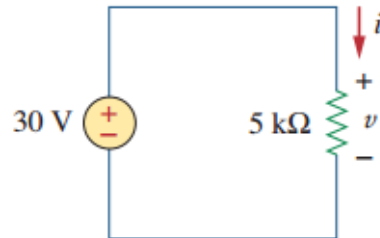


Figure 2.8
For Example 2.2.

In the circuit shown in Fig. 2.8, calculate the current i , the conductance G , and the power p .

Solution:

The voltage across the resistor is the same as the source voltage (30 V) because the resistor and the voltage source are connected to the same pair of terminals. Hence, the current is

$$i = \frac{v}{R} = \frac{30}{5 \times 10^3} = 6 \text{ mA}$$

The conductance is

$$G = \frac{1}{R} = \frac{1}{5 \times 10^3} = 0.2 \text{ mS}$$

The power is

$$p = vi = 30 \text{ V} \times 6 \text{ mA} = 180 \text{ mW}$$

or

$$p = v^2 G = 30^2 \times 0.2 = 180 \text{ mW}$$

or

$$p = i^2 R = (6e^{-3})^2 \times 5e^3 = 0.18 \text{ W}$$

Outline

- Ohm's Law
- **Terminology**
 - Branches, Nodes, Loops, Meshes
- **Kirchhoff's Law**
- Series Resistors and Voltage Division
- Parallel Resistors and Current Division
- Wye-Delta Transformations

Terminology (Branches and Nodes)

- An electric circuit is an interconnection of electrical elements
 - How the elements are interconnected in a circuit is very important. Some knowledge about topology are required
- A **branch (支路)** represents any single element in a circuit, such as a voltage source or a resistor
 - 5 branches in Fig. 2.10: the 10-V source, the 2-A source, and the three resistors
- A **node (节点)** is the point of connection between two or more branches
 - 3 nodes: a, b, c

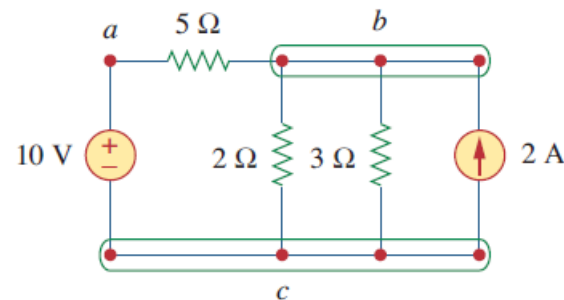


Figure 2.10
Nodes, branches, and loops.

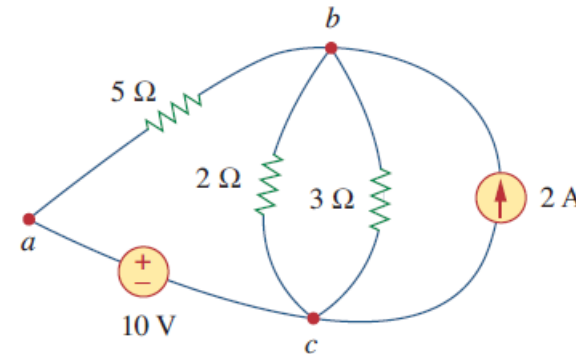


Figure 2.11
The three-node circuit of Fig. 2.10 is redrawn.

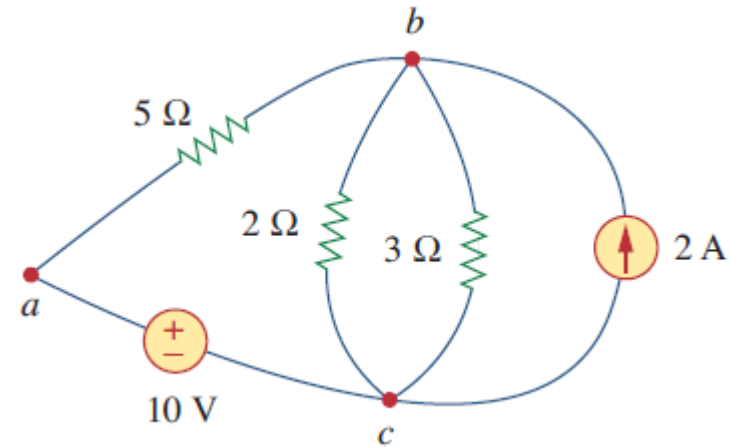
Terminology (Loops and Meshes)

- A **loop** (回路) is any closed path in a circuit (*e.g.*, 6 loops)
- A set of loops is said **independent** (独立回路) when any of them contains at least one new branch in addition to all others (*e.g.*, 3 independent loops)
- A **mesh** (网孔) is a loop which does not contain any other loops within it (*e.g.*, 3)

- Theorem of network topology

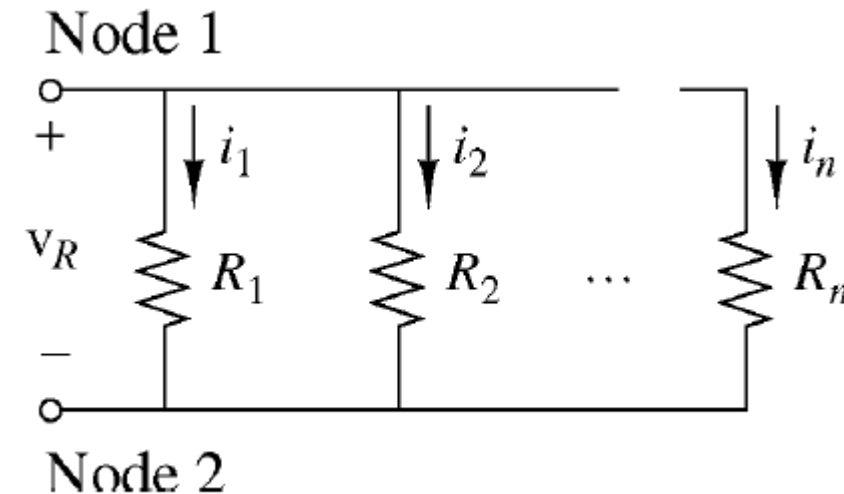
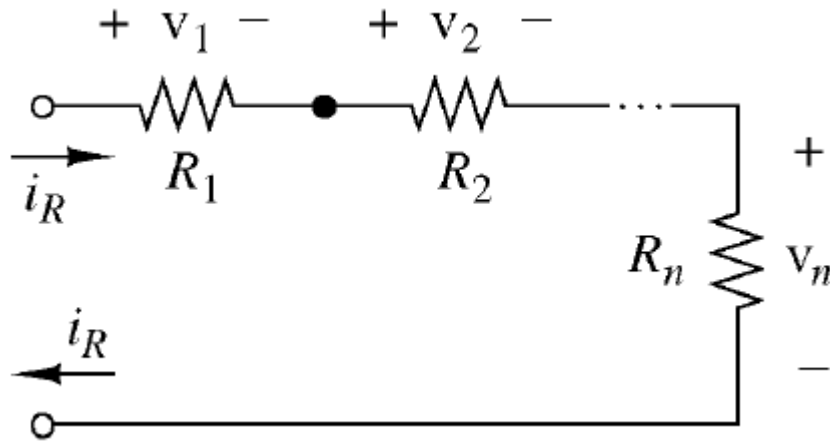
$$b = l + n - 1$$

- **b**: number of **b**ranches
- **l**: number of independent **l**oops
- **n**: number of **n**odes

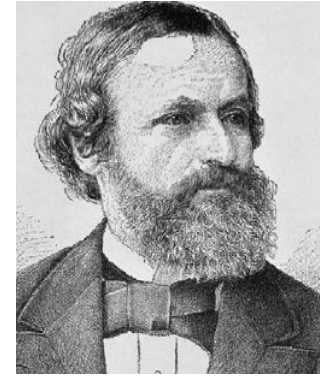


Series and Parallel Connections

- Two or more elements are connected “**in series (串联)**” if they are serially connected and thus carry the same current
- Two or more elements are “**in parallel (并联)**” if they are connected to the same two nodes and thus have the same voltage across them



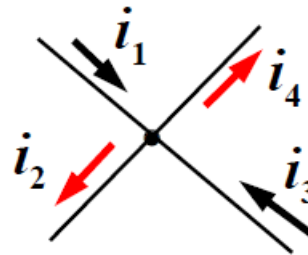
Kirchhoff's Current Law (KCL)



- Gustav Robert Kirchhoff (1824-1887), a German physicist, stated two basic laws concerning the relationship between the currents and voltages in an electrical circuit
- KCL (基尔霍夫电流定律):** The algebraic sum of the currents entering a node (or a closed boundary) is zero. In other words, the sum of the currents entering a node is equal to that leaving the node

$$\sum_{n=1}^N i_n = 0$$

$$\sum i_{\text{in}} = \sum i_{\text{out}}$$

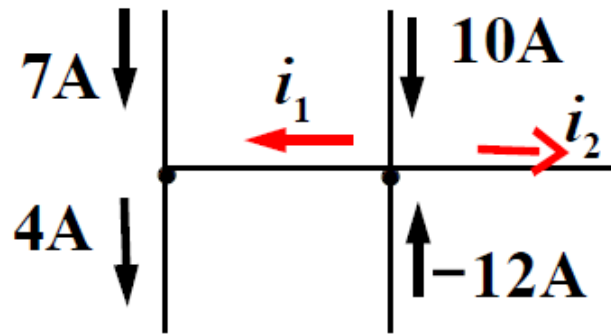


$$-i_1 + i_2 - i_3 + i_4 = 0$$

$$i_1 + i_3 = i_2 + i_4$$

Example: KCL

- The following network is represented by its topological diagram. Find the unknown currents in the network



$$4 - 7 - i_1 = 0$$

$$4 = 7 + i_1$$



$$i_1 = -3\text{A}$$

$$i_1 + i_2 - 10 - (-12) = 0$$

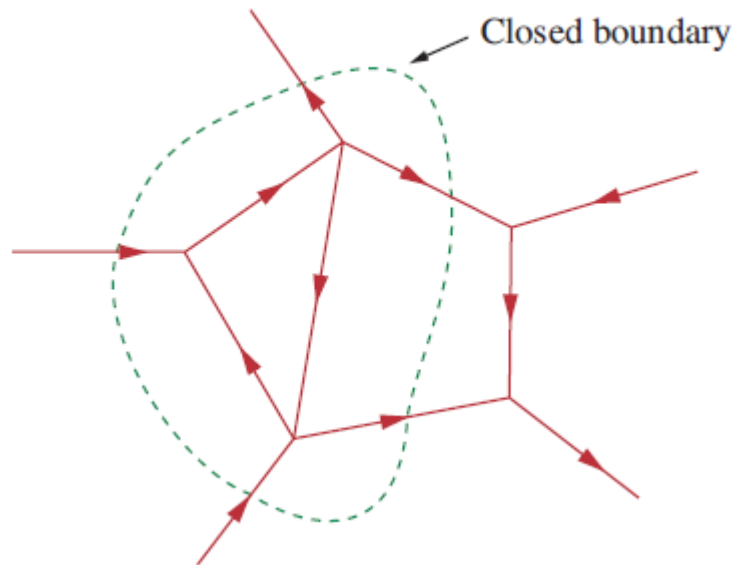
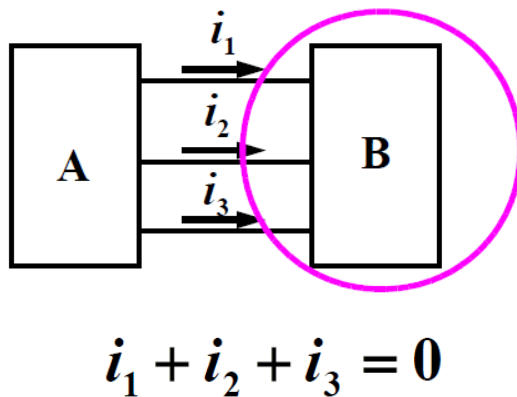
$$i_1 + i_2 = 10 - 12$$



$$i_2 = 1\text{A}$$

Closed Boundary (闭合面)

- KCL is based on the law of conservation of charge
 - The algebraic sum of charges within a system cannot change (true under the lumped circuit assumption/abstraction)
- The KCL also applies to a closed boundary → Regarded as a generalized case
- A closed boundary is a closed curve (or surface), such as a circle in a plane that has a well-defined inside and outside. This closed boundary is sometimes called a Gauss surface (高斯面)



Johann Carl Friedrich Gauss
 (1777-1855)
 German mathematician

Kirchhoff's Voltage Law (KVL)

- KVL (基尔霍夫电压定律):** The algebraic sum of the voltage drops around any closed path (or loop) is zero at any instance of time

$$\sum_{m=1}^M v_m = 0$$

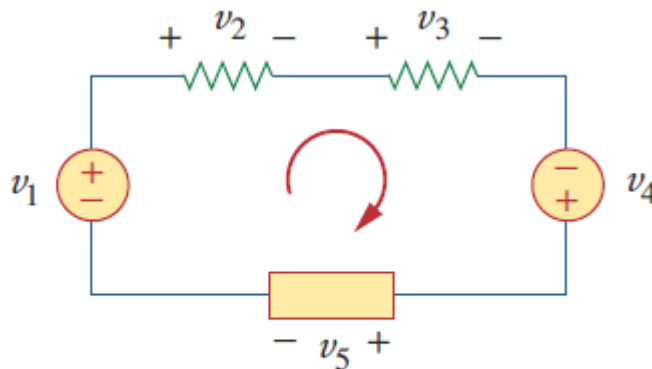


Figure 2.19

A single-loop circuit illustrating KVL.

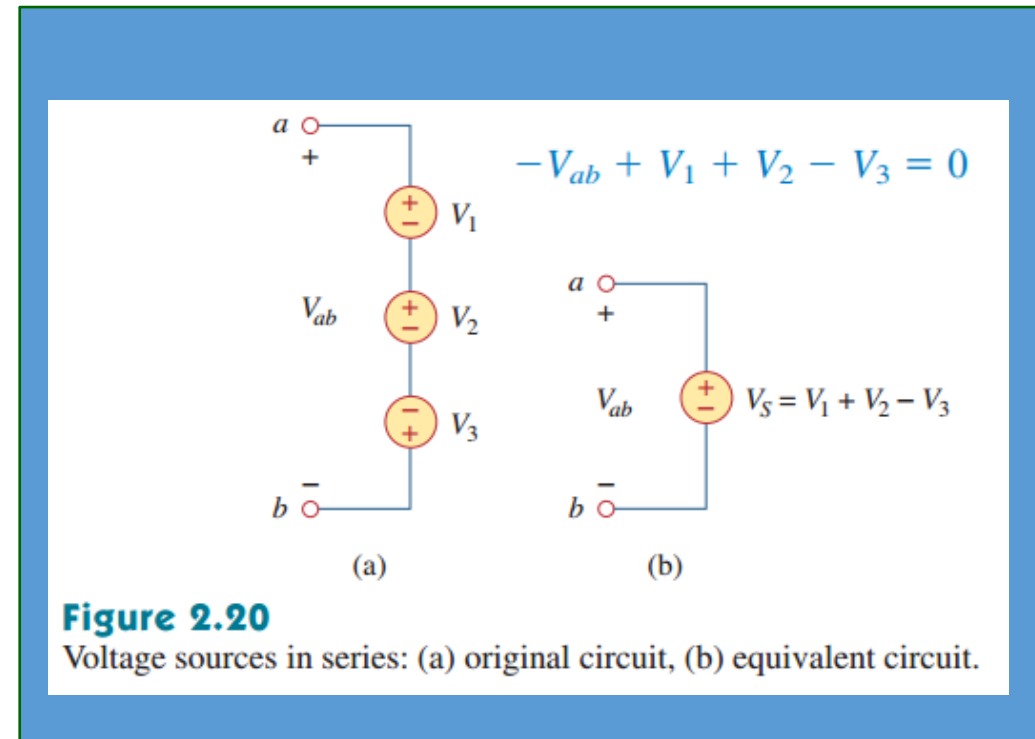
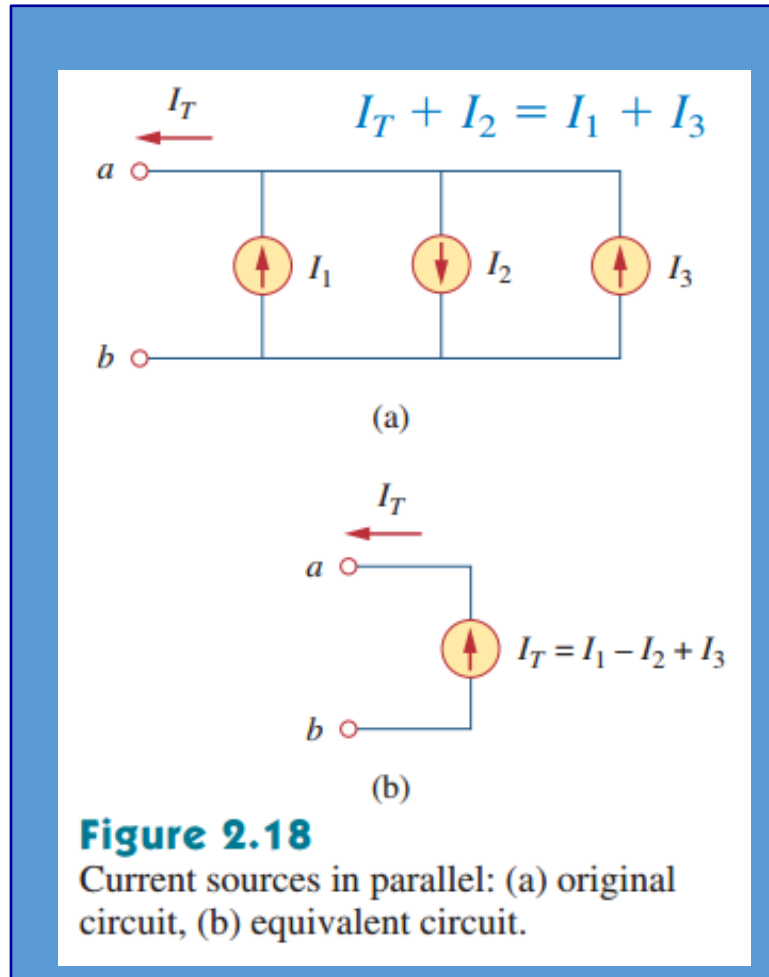
$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

$$v_2 + v_3 + v_5 = v_1 + v_4$$

Sum of voltage drops = Sum of voltage rises

Equivalent Circuits/Sources

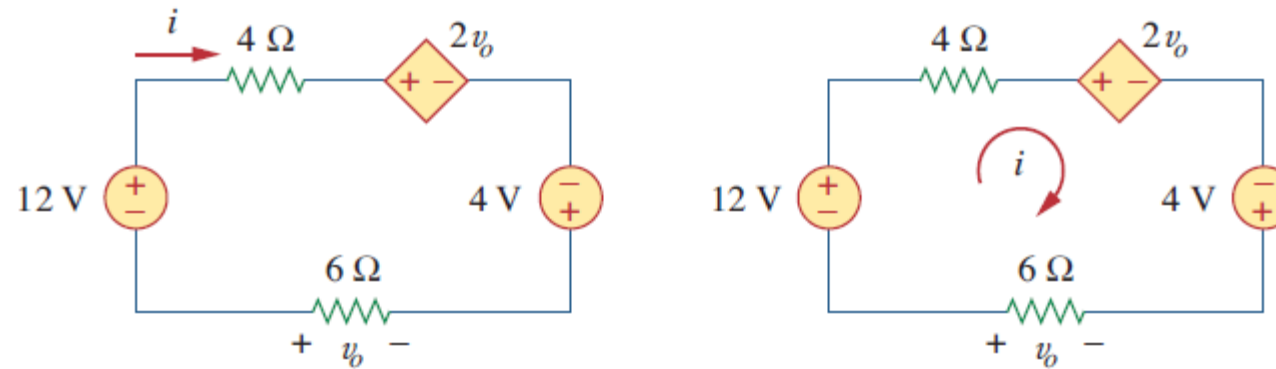
- Two circuits (or sources) are said to be **equivalent (等效的)** if they have the same **V-I relationship (伏安特性)** at their terminals (端口)



Example: KVL

Example 2.6

Determine v_o and i in the circuit



Solution:

We apply KVL around the loop as shown in Fig. 2.23(b). The result is

$$-12 + 4i + 2v_o - 4 + 6i = 0 \quad (2.6.1)$$

Applying Ohm's law to the 6-Ω resistor gives

$$v_o = -6i \quad (2.6.2)$$

Substituting Eq. (2.6.2) into Eq. (2.6.1) yields

$$-16 + 10i - 12i = 0 \quad \Rightarrow \quad i = -8 \text{ A}$$

and $v_o = 48 \text{ V}$.

Example: KCL & KVL

Example 2.8

Find currents and voltages in the circuit shown in Fig. 2.27(a).

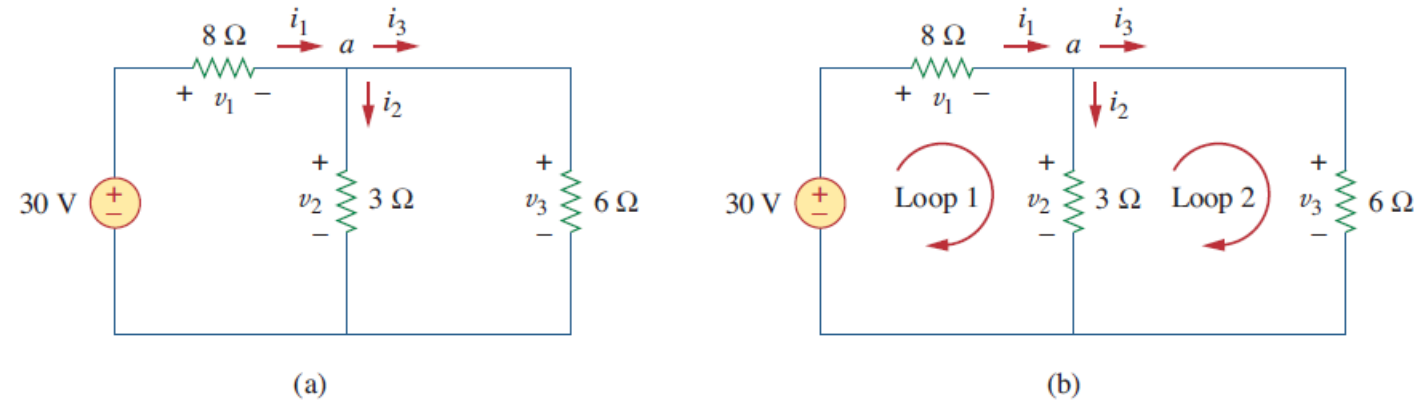


Figure 2.27

Solution:

KCL at node a: $i_1 - i_2 - i_3 = 0$;

KVL law for loop 1: $-30 + v_1 + v_2 = 0$; KVL law for loop 2: $-v_2 + v_3 = 0$

Ohm's law, $v_1 = 8i_1$; $v_2 = 3i_2$; $v_3 = 6i_3$;

Solve the above equations, we get:

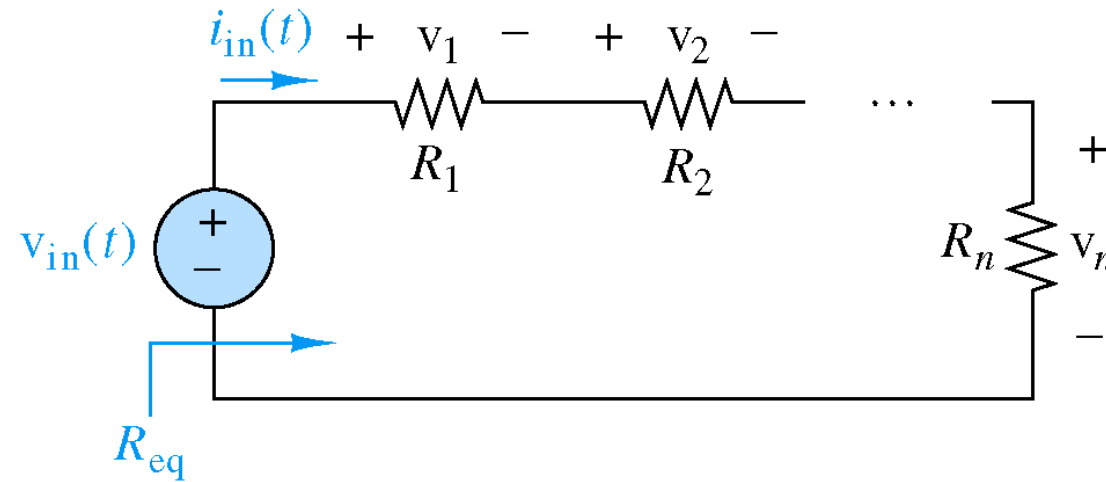
$i_1 = 3\text{A}$, $i_2 = 2\text{A}$, $i_3 = 1\text{A}$; $v_1 = 24\text{V}$, $v_2 = 6\text{V}$, $v_3 = 6\text{V}$.

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Series Resistors

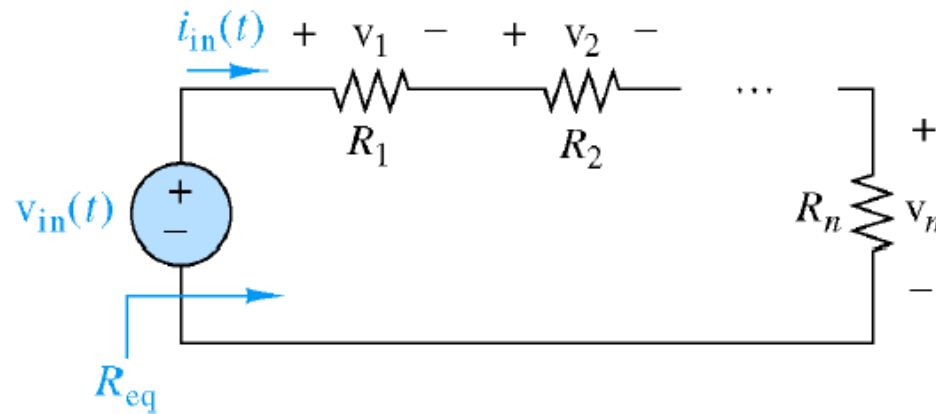
- The equivalent resistance of any number of resistors connected in series is the sum of the resistors (Why?)



$$R_{eq} = R_1 + R_2 + \cdots + R_n$$

Voltage Division

- In a series combination of n resistors, the voltage drop across the resistor R_j for $j = 1, 2, \dots, n$ is

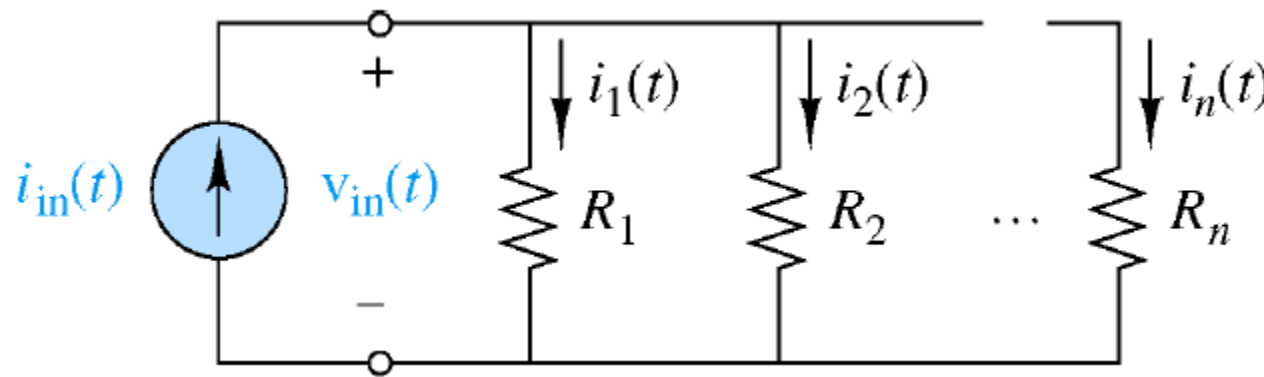


$$v_j(t) = \frac{R_j}{R_1 + R_2 + \dots + R_n} v_{in}(t)$$

- This is the principle of **voltage division** (分压)

Parallel Resistors

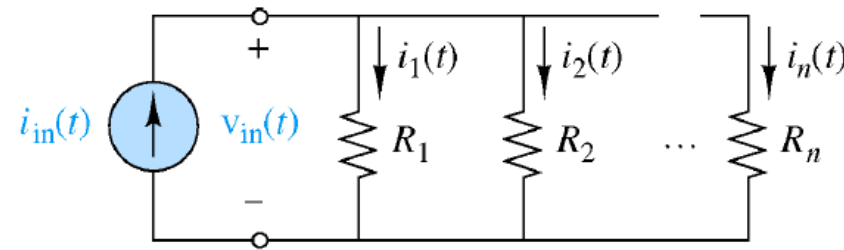
- The equivalent conductance of resistors connected in parallel is the sum of their individual conductances (Why?)



$$G_{eq} = G_1 + G_2 + \dots + G_n \quad \text{or} \quad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

Current Division

- In a parallel combination of n resistors, the current through the resistor R_j for $j = 1, 2, \dots, n$ is:



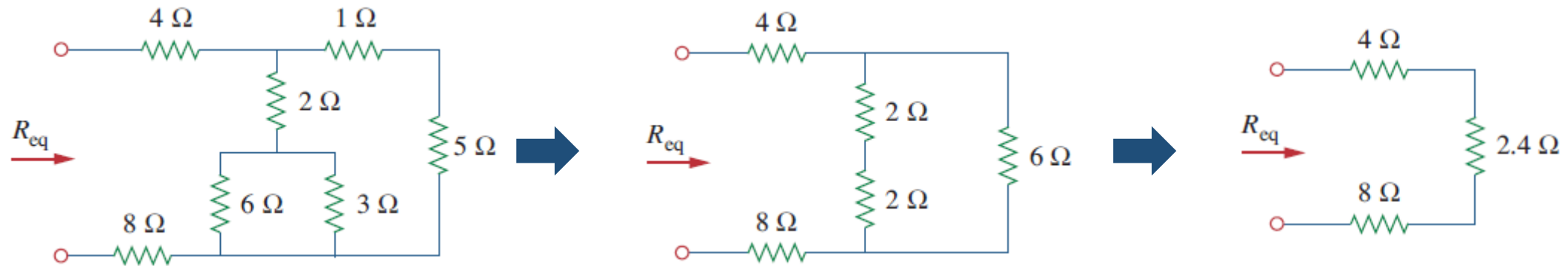
$$i_j(t) = \frac{G_j}{G_1 + G_2 + \dots + G_n} i_{in}(t)$$

- This is the principle of **current division (分流)**
- Q: What is the formula for two parallel resistors?!*

$$i_1 = \frac{R_2 i}{R_1 + R_2}, \quad i_2 = \frac{R_1 i}{R_1 + R_2}$$

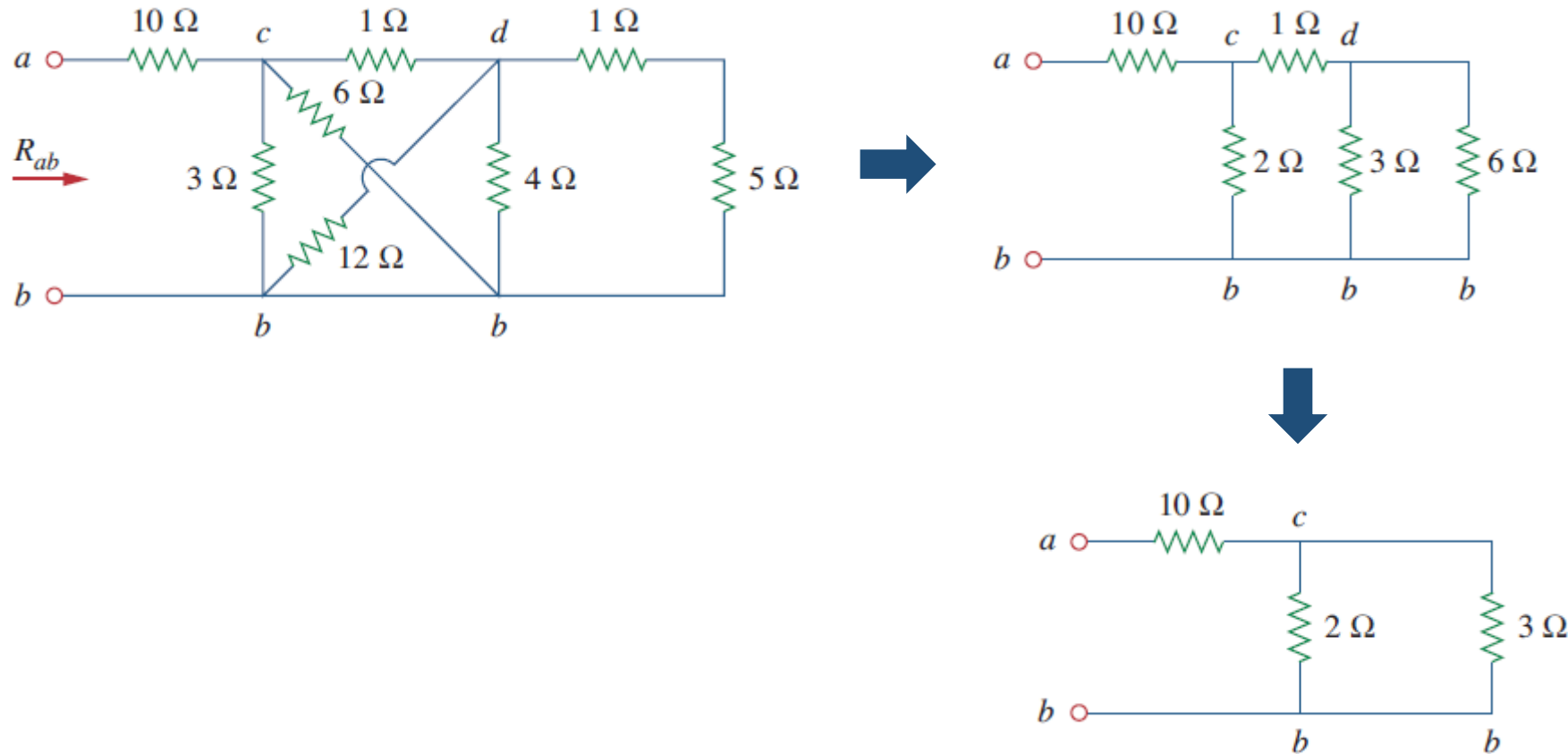
Example: Series and Parallel Resistors

- In the following circuit find R_{eq}



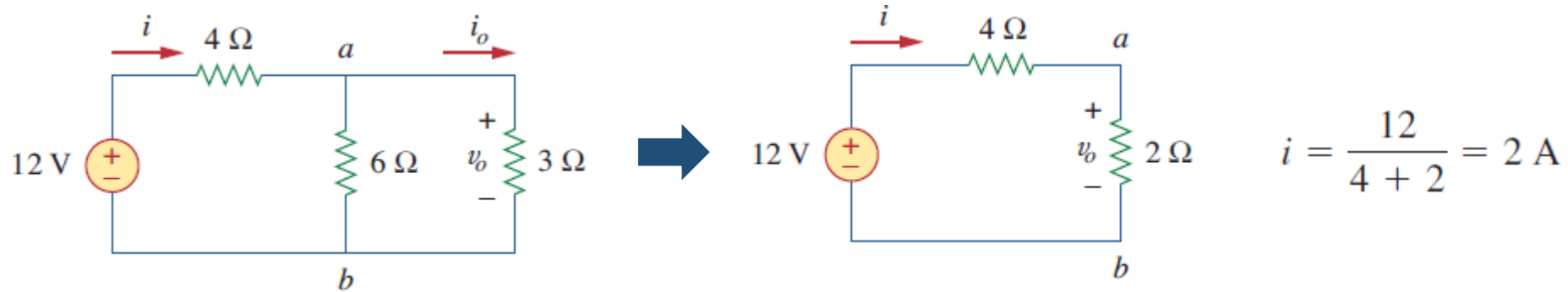
Example: Series and Parallel Resistors

- Calculate the equivalent R_{eq} in the following circuit



Example: Series and Parallel Resistors

- In the following circuit find i_o and v_o

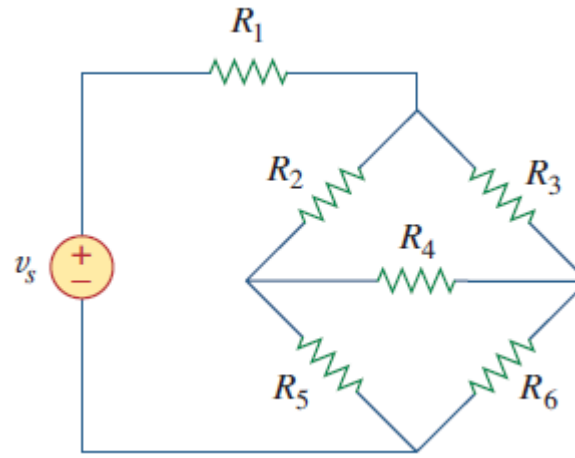


$$v_o = \frac{2}{2 + 4} (12\text{ V}) = 4\text{ V}$$

$$v_o = 3i_o = 4 \quad \Rightarrow \quad i_o = \frac{4}{3}\text{ A}$$

Wye-Delta Transformations

- In some circuits the resistors are neither in series nor in parallel
- For example, consider the following bridge circuit



- How can we combine the resistors R_1 through R_6 ?

Wye and Delta Networks

- A useful technique that can be used to simplify many such circuits is transformation from wye (Y) to delta (Δ) network
- A wye (Y) or tee (T) network
- The delta (Δ) or pi (π) network

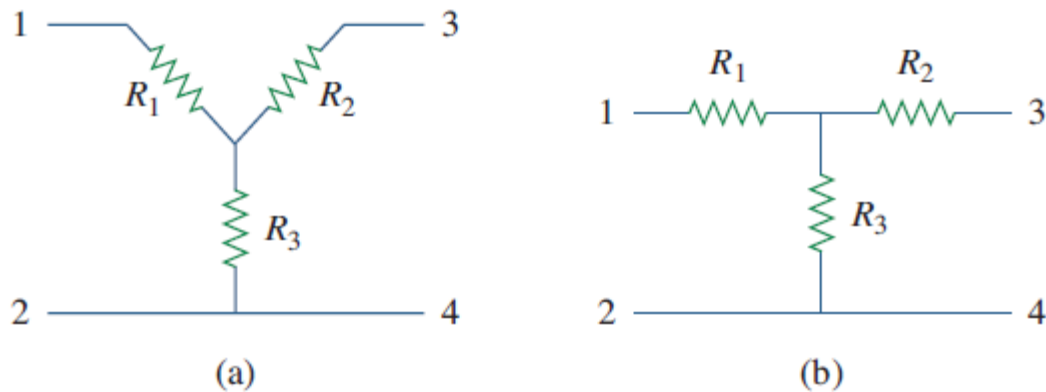


Figure 2.47

Two forms of the same network: (a) Y, (b) T.

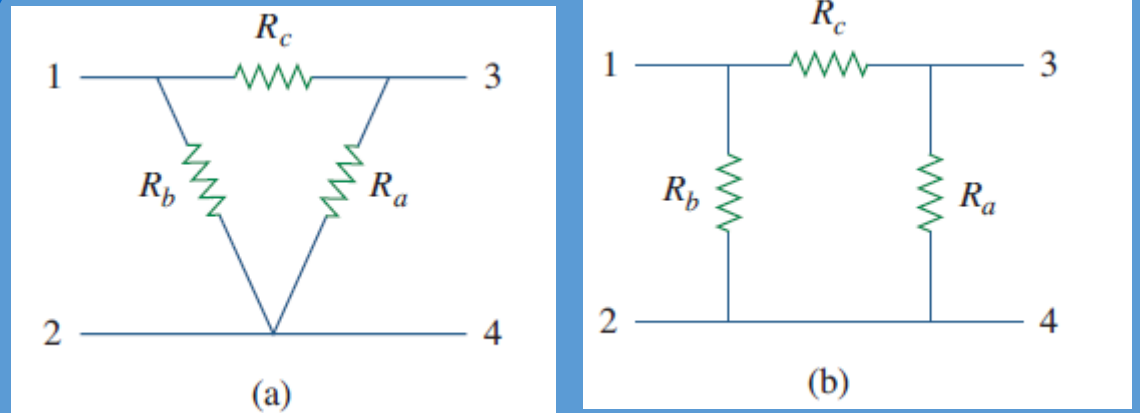


Figure 2.48

Two forms of the same network: (a) Δ , (b) Π .

Delta-Wye Conversion (I)

- In some cases, it is more convenient to work with a Y network in place of a Δ network
- Let's superimpose a wye network on the existing delta network and try to find the equivalent resistances in the wye network
- We calculate the equivalent resistance between terminals **1** and **2**

$$R_{12}(Y) = R_1 + R_3$$

$$R_{12}(\Delta) = R_b \parallel (R_a + R_c)$$

Setting $R_{12}(Y) = R_{12}(\Delta)$ gives

$$R_{12} = R_1 + R_3 = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c}$$

Similarly

$$R_{13} = R_1 + R_2 = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c}$$

$$R_{34} = R_2 + R_3 = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c}$$

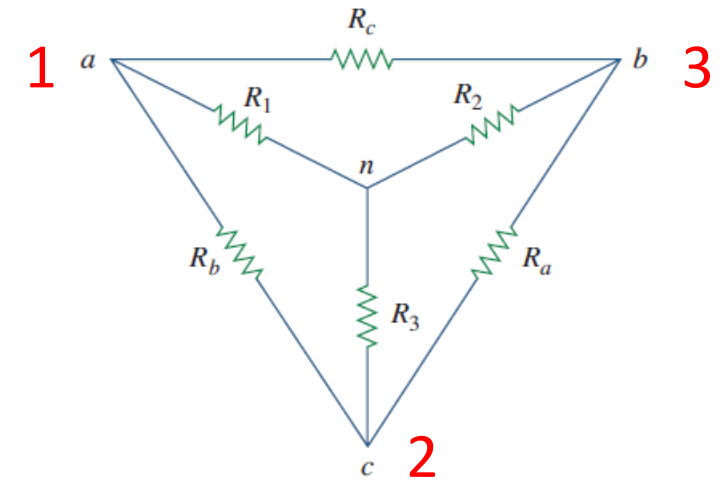
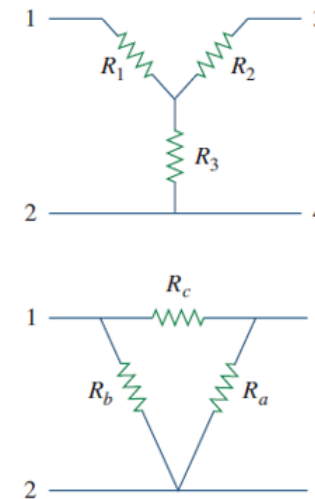


Figure 2.49
Superposition of Y and Δ networks as an aid in transforming one to the other.

Delta-Wye Conversion (II)

- Solving for R_1 , R_2 , and R_3 we have

$$\begin{aligned}
 R_{12} = R_1 + R_3 &= \frac{R_b(R_a + R_c)}{R_a + R_b + R_c} \\
 R_{13} = R_1 + R_2 &= \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} \\
 R_{34} = R_2 + R_3 &= \frac{R_a(R_b + R_c)}{R_a + R_b + R_c}
 \end{aligned}
 \rightarrow$$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

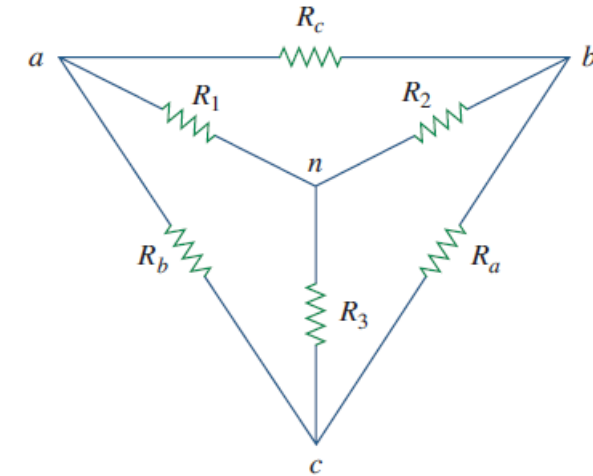


Figure 2.49

Superposition of Y and Δ networks as an aid in transforming one to the other.

- Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches, divided by the sum of the three Δ resistors
- Δ network is kind of like parallel connection, $R_a \rightarrow 0$, $R_1 = R_b // R_c$
- So, R_Y is less than R_Δ

Wye-Delta Conversion

$$\Delta \text{ to Y: } R_1 = \frac{R_b R_c}{R_a + R_b + R_c}; \quad R_2 = \frac{R_a R_c}{R_a + R_b + R_c}; \quad R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$\text{Y to } \Delta: \left\{ \begin{aligned} R_a &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} = R_2 + R_3 + \frac{R_2 R_3}{R_1} \\ R_b &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} = R_1 + R_3 + \frac{R_1 R_3}{R_2} \\ R_c &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} = R_1 + R_2 + \frac{R_1 R_2}{R_3} \end{aligned} \right.$$

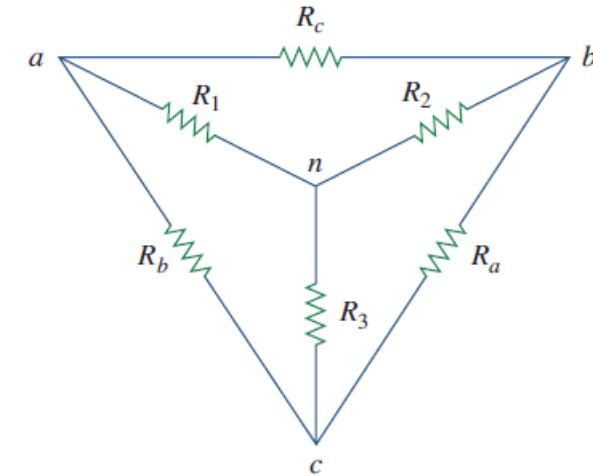


Figure 2.49

Superposition of Y and Δ networks as an aid in transforming one to the other.

- Each resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor
- The Y-network is more like a serial connection while the Δ -network like a “parallel” connection. So, R_Y is less than R_Δ

Example: Delta-Wye Conversion

Convert the Δ network in Fig. 2.50(a) to an equivalent Y network.

Example 2.14

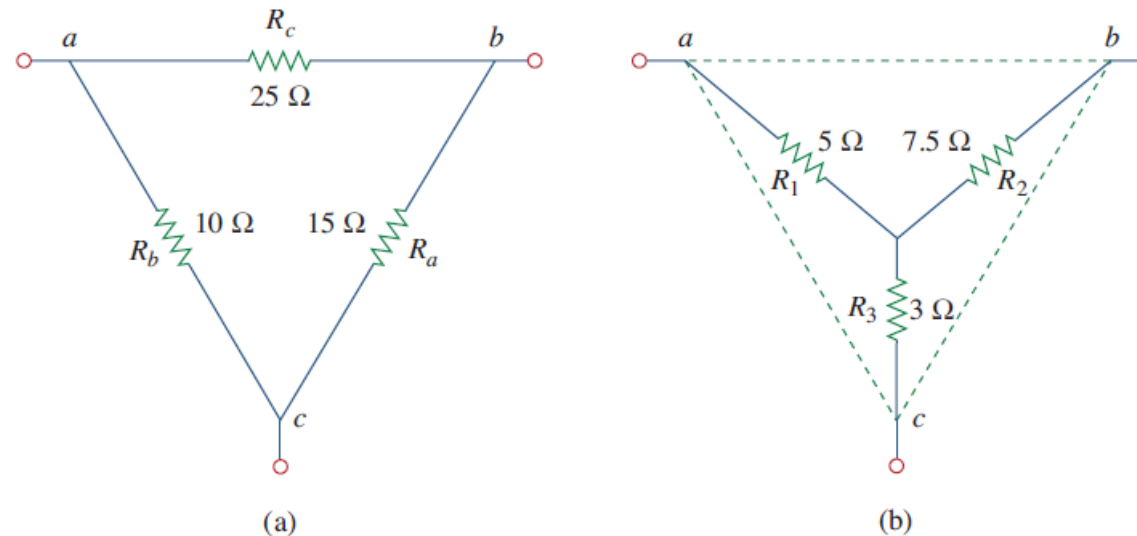


Figure 2.50

For Example 2.14: (a) original Δ network, (b) Y equivalent network.

Solution:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{10 \times 25}{15 + 10 + 25} = \frac{250}{50} = 5 \, \Omega$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} = \frac{25 \times 15}{50} = 7.5 \, \Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{15 \times 10}{50} = 3 \, \Omega$$

Example: Wye-Delta Conversion

Example 2.15

Obtain the equivalent resistance R_{ab} for the circuit in Fig. 2.52 and use it to find current i .

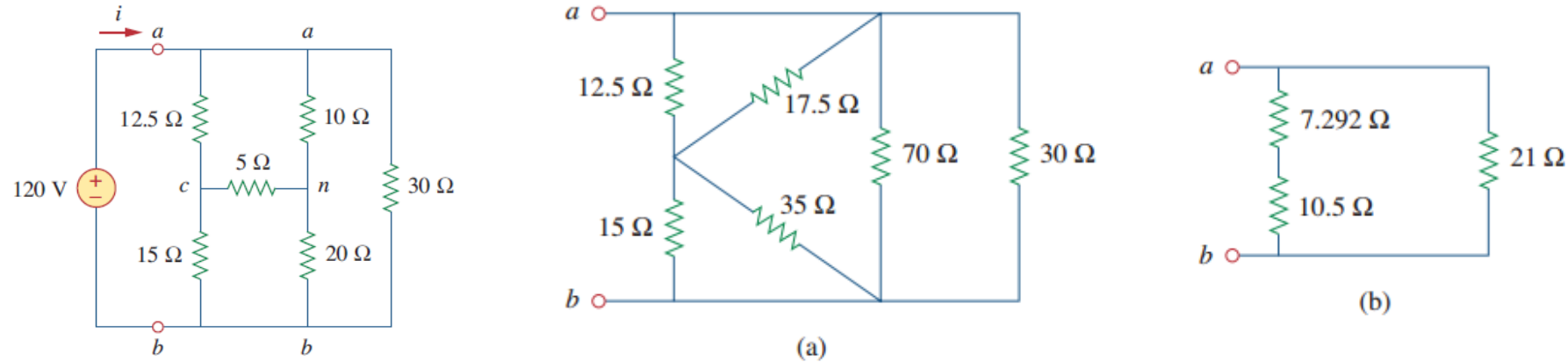


Figure 2.52
For Example 2.15.

Solution:

Applying Y to Δ transformation, we get

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} = \frac{10 \times 20 + 20 \times 5 + 5 \times 10}{10}$$

$$= \frac{350}{10} = 35 \Omega$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} = \frac{350}{20} = 17.5 \Omega$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} = \frac{350}{5} = 70 \Omega$$

Thus, $12.5 \parallel 17.5 = \frac{12.5 \times 17.5}{12.5 + 17.5} = 7.292 \Omega$

$$15 \parallel 35 = \frac{15 \times 35}{15 + 35} = 10.5 \Omega$$

$$R_{ab} = (7.292 + 10.5) \parallel 21 = \frac{17.792 \times 21}{17.792 + 21} = 9.632 \Omega$$

$$i = \frac{v_s}{R_{ab}} = \frac{120}{9.632} = 12.458 \text{ A}$$

Summary (I)

- A resistor is a passive element in which the voltage v across it is directly proportional to the current i through it. $v = iR$
- The conductance G of a resistor is the reciprocal of its resistance: $G = \frac{1}{R}$
- Kirchhoff's current law (KCL) states that the currents at any node algebraically sum to zero $\sum_{n=1}^N i_n = 0$
- Kirchhoff's voltage law (KVL) states that the voltages around a closed path algebraically sum to zero $\sum_{m=1}^M v_m = 0$
- Two elements are in series when they are connected sequentially, end to end. When elements are in series, the same current flows through them ($i_1 = i_2$). They are in parallel if they are connected to the same two nodes. Elements in parallel always have the same voltage across them ($v_1 = v_2$).
- When two resistors $R_1 (=1/G_1)$ and $R_2 (=1/G_2)$ are in series, their equivalent resistance R_{eq} and equivalent conductance G_{eq} are

$$R_{eq} = R_1 + R_2, \quad G_{eq} = \frac{G_1 G_2}{G_1 + G_2}$$

Summary (II)

- When two resistors $R_1 (=1/G_1)$ and $R_2 (=1/G_2)$ are in parallel, their equivalent resistance R_{eq} and equivalent conductance G_{eq} are

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}, \quad G_{eq} = G_1 + G_2$$

- The voltage division principle for two resistors in series is $v_1 = \frac{R_1}{R_1 + R_2} v, \quad v_2 = \frac{R_2}{R_1 + R_2} v$
- The current division principle for two resistors in parallel is $i_1 = \frac{R_2}{R_1 + R_2} i, \quad i_2 = \frac{R_1}{R_1 + R_2} i$
- The formulas for a delta-to-wye transformation are

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}, \quad R_2 = \frac{R_c R_a}{R_a + R_b + R_c}, \quad R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

- The formulas for a wye-to-delta transformation are

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}, \quad R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}, \quad R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

- Acknowledgement: This Lecture Note is partially based on materials from Prof. Chenchang Zan and partially adapted from the notes of the book “Fundamentals of Electric Circuits” by C. K. Alexander and M. N. O. Sadiku